

1. Linearly separable:

- A. in 2D space, the data points can be separated by a straight line.
- B. in 3D space, the data points can be separated by a flat plane.

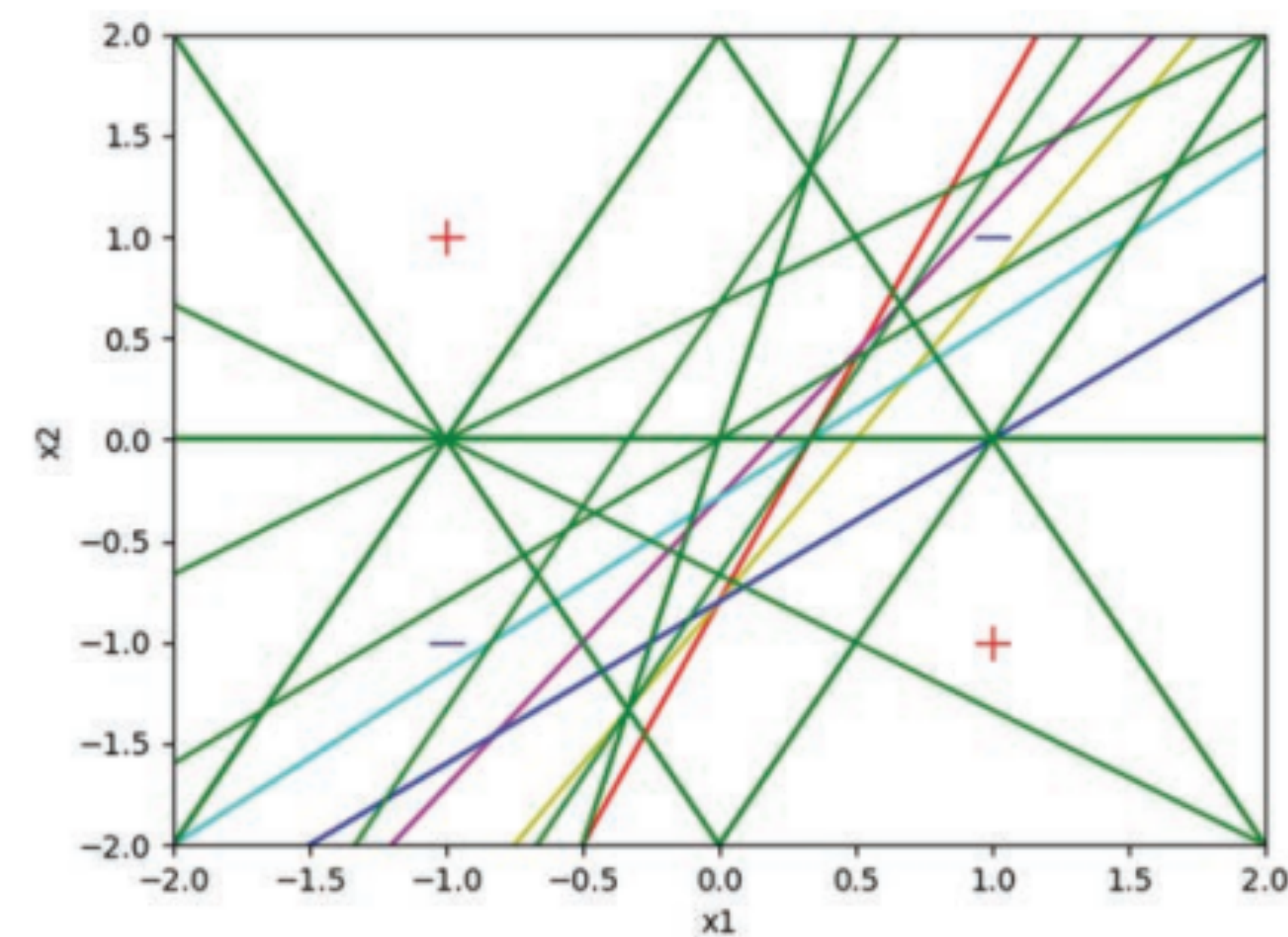
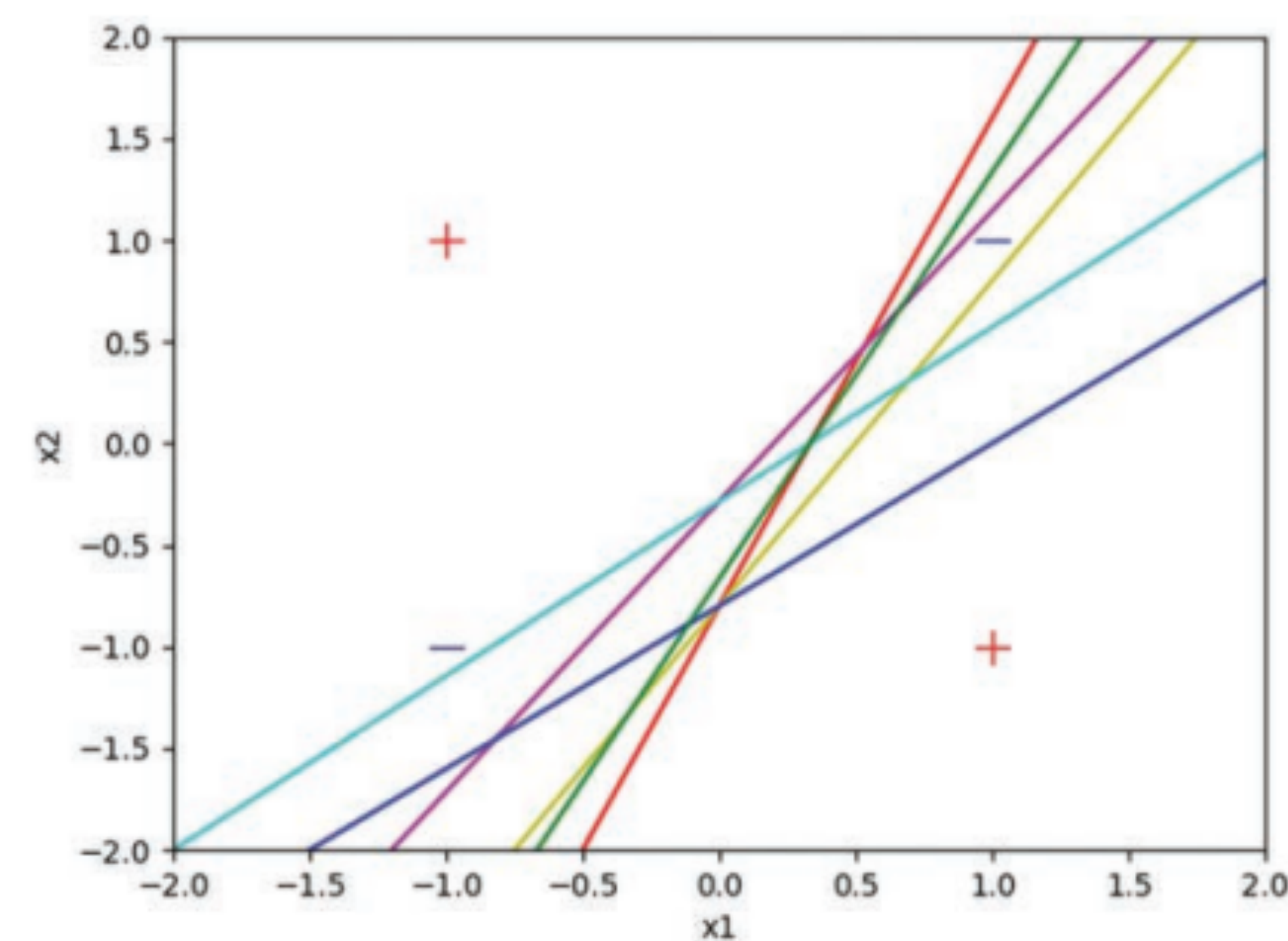
2. limitation of a single perceptron:

- A. a single perceptron can only solve linearly separable issues.
- B. it's impossible to separate XOR Gate data points with a straight line in 2D space, but several straight lines is available.
- C. in other words, it's impossible to solve XOR Gate issues with a single perceptron, but combining multiple perceptrons is available.

D.

**Table 1-2** Truth Table for a Two-Input XOR Gate

| $x_0$ | $x_1$ | $y$   |
|-------|-------|-------|
| False | False | False |
| True  | False | True  |
| False | True  | True  |
| True  | True  | False |



**Figure 1-7** Perceptron attempting to learn XOR. Top: After 6 weight adjustments. Bottom: After 30 weight adjustments.

the result above shows that a single perceptron using perceptron learning algorithm never make the perceptron learning algorithm converge in a XOR Gate issue

3. learning algorithm converge: learning algorithm successfully finds out fixed set of weights for perceptron(s) to correctly match pair training example(s).